

Sampling in Action: The M&M Challenge

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Roadmap

- + M&M Sampling Activity
- + Analysis and Discussion
- + Advanced Sampling Concepts

What is Sampling?

- + Selecting a subset of a population
- + Used to estimate characteristics of the whole
- + Critical in research and statistics

M&M Sampling Activity

- + Objective: Demonstrate sampling principles using M&M's
 - + Hands-on experience with data collection and analysis
- + Materials:
 - + Small packages of plain M&M's (one per student)
 - + Napkins for sorting

M&M Sampling Procedure

Steps in the activity

- + Distribute M&M packages and materials
- + Sort M&M's by color on napkins
- + Record frequency of each color
- + Calculate percentages for each color
- + Hypothesize population color distribution
- + Form pairs to pool data
- + Pool data for entire class
 - + using google sheets (and some R magic)

Data Collection



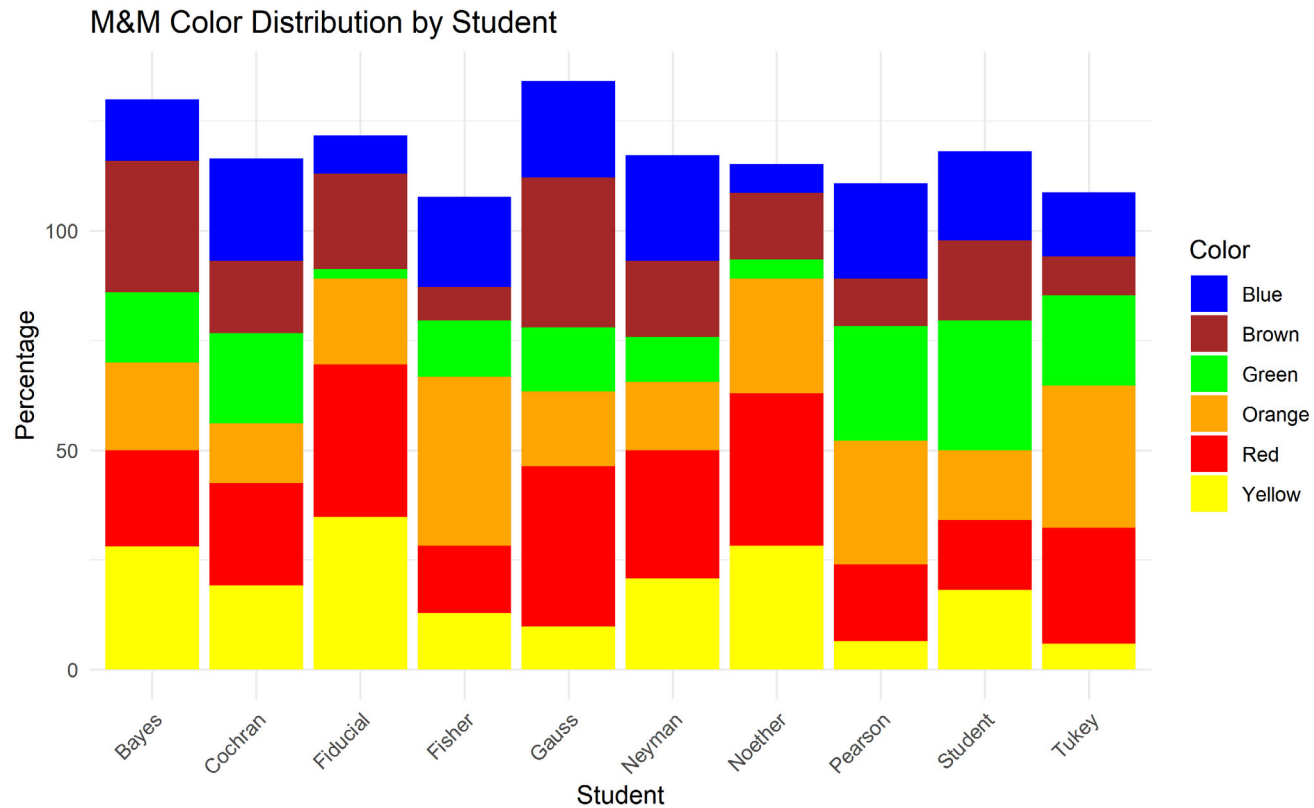
Scan to input your data!

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https://docs.google.com/spreadsheets/d/1D4i8e0pTrqwLk_FjMFtkirX6OU9RT57m_Q

Analysis in Action

+ What we'll get from the class data



Source Code

```
set.seed(123) # For reproducibility

# Define the number of students and colors
students <- c("Tukey", "Gauss", "Noether", "Fisher",
             "Bayes", "Pearson", "Student",
             "Fiducial", "Neyman", "Cochran")
colors <- c("Blue", "Orange",
           "Green", "Red", "Yellow", "Brown")

# Simulate the total number of M&Ms for each student
total_mms <- sample(15:20,
                   length(students), replace = TRUE)

# Simulate the counts of each color for each student
color_counts <- replicate(length(colors),
                          sample(1:total_mms,
                                length(students), replace = TRUE))

# Create the dataframe
df_syn <- data.frame(Name = students,
                    color_counts)
colnames(df_syn)[-1] <- colors

# Calculate the percentages
df_syn <- df_syn %>%
  mutate(Total = rowSums(across(Blue:Yellow))) %>%
  mutate(Blue_perc = Blue / Total * 100,
         Orange_perc = Orange / Total * 100,
         Green_perc = Green / Total * 100,
         Red_perc = Red / Total * 100,
         Brown_perc = Brown / Total * 100,
         Yellow_perc = Yellow / Total * 100)

# Reshape the data to long format
df_long_syn <- df_syn %>%
  pivot_longer(cols = c(Blue_perc, Orange_perc, Green_perc, Red_perc, Yellow_perc, Brown_perc),
              names_to = "Color",
              values_to = "Percentage")

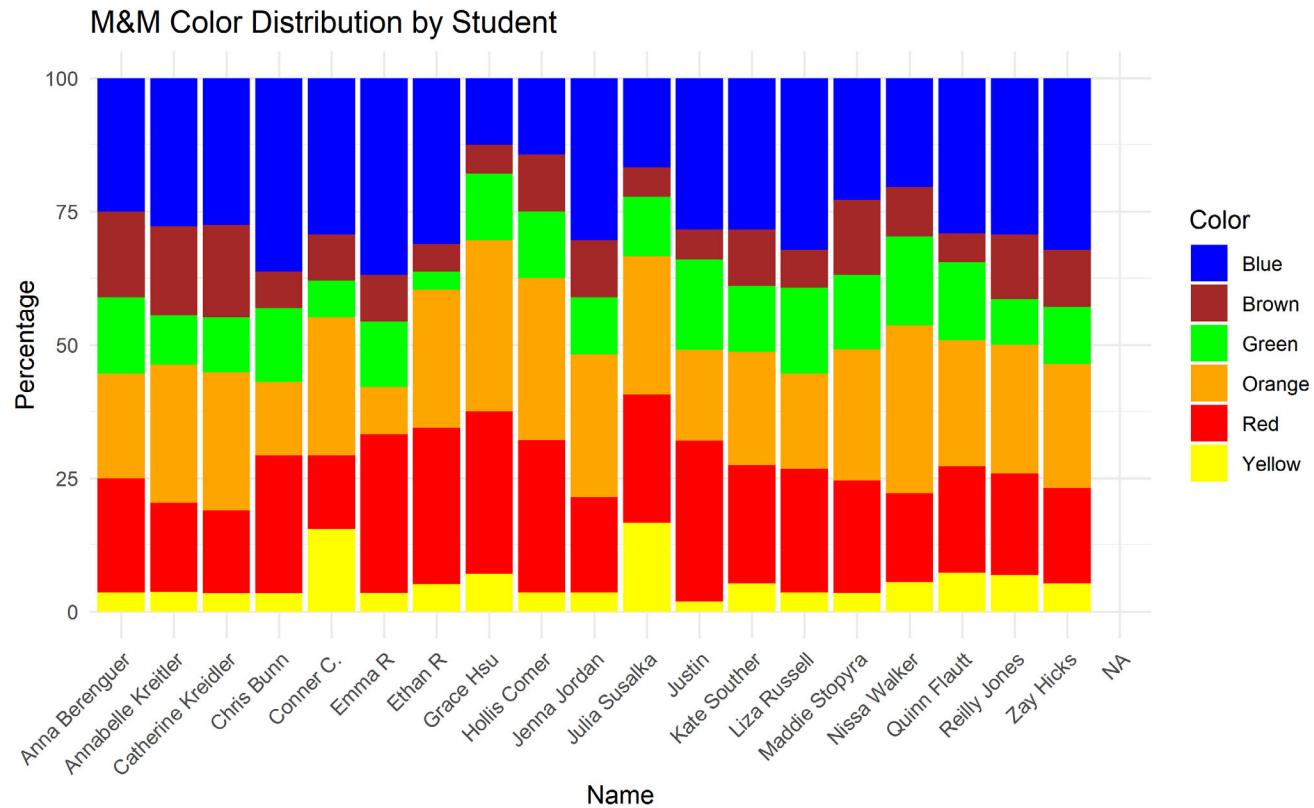
# Plotting the data
stacked_plot <- df_long_syn %>%
  ggplot(aes(x = Name, y = Percentage, fill = Color)) +
  geom_col(position = "stack") +
  labs(title = "M&M Color Distribution by Student",
       x = "Student",
       y = "Percentage") +
  scale_fill_manual(values = c("Blue_perc" = "blue",
                              "Orange_perc" = "orange",
                              "Green_perc" = "green",
                              "Red_perc" = "red",
                              "Yellow_perc" = "yellow",
                              "Brown_perc" = "brown"),
                   labels = c("Blue_perc" = "Blue",
                              "Orange_perc" = "Orange",
                              "Green_perc" = "Green",
                              "Red_perc" = "Red",
                              "Brown_perc" = "Brown",
                              "Yellow_perc" = "Yellow")) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))

# Overall distribution of M&Ms
overall_distribution <- df_syn %>%
  select(Blue, Orange, Green, Red, Yellow) %>%
  summarise(across(everything(), sum)) %>%
  pivot_longer(cols = everything(), names_to = "Color", values_to = "Count")

overall_plot <- overall_distribution %>%
  ggplot(aes(x = Color, y = Count, fill = Color)) +
  geom_col() +
  labs(title = "Overall M&M Color Distribution",
       x = "Color",
       y = "Total Count") +
  scale_fill_manual(values = c("Blue" = "blue",
                              "Orange" = "orange",
                              "Green" = "green",
                              "Red" = "red",
                              "Yellow" = "yellow",
                              "Brown" = "brown")) +
  theme_minimal()

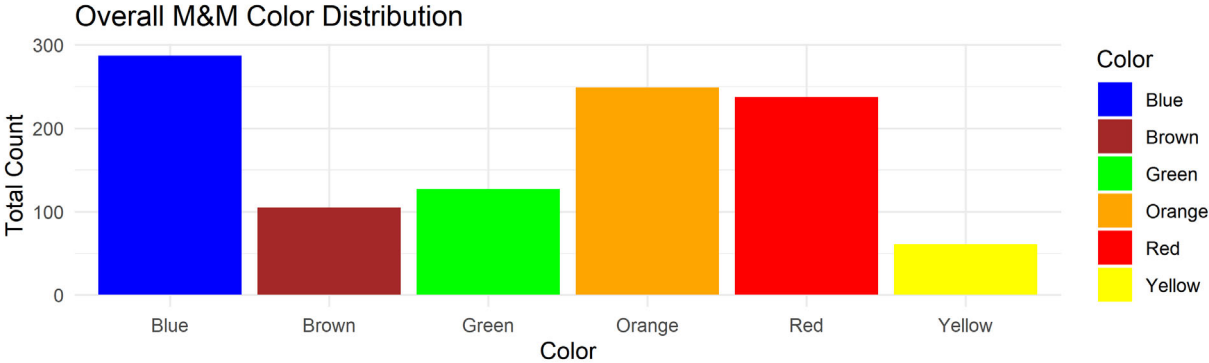
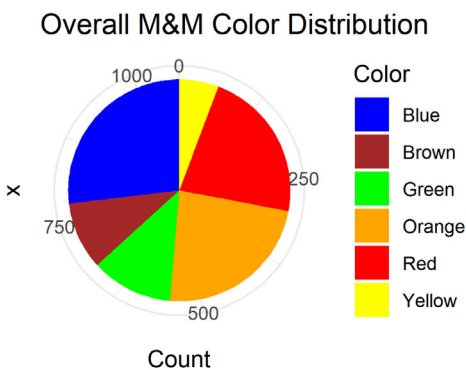
stacked_plot
```

Analysis in action



Sample Size Effects

Sample.Size	Accuracy	Variability
Individual	Low	High
Paired	Medium	Medium
Class-wide	High	Low



Advanced Sampling Concepts

Relating to Sampling Methods

- + Simple random sampling
 - + Each M&M package as a random sample
- + Stratified sampling
 - + If we sorted M&M bags by production date,
 - + Could this improve representativeness?
- + Cluster sampling
 - + If we sampled entire boxes of M&M packages
 - + Potential production batch effects?
- + Systematic sampling
 - + If we selected every n th M&M package from production line
 - + Could introduce cyclical biases?

Potential Biases in M&M Sampling



Production process biases

- + Color distribution variations between factories
- + Akin to sampling frame bias in surveys



Selection bias

- + If students choose their favorite color of package
- + Akin to non-random sample selection in research



Measurement bias

- + Errors in counting or recording M&M colors
- + Akin to survey response errors



Non-response bias

- + If some students don't participate or eat their M&M's
- + Akin to survey non-respondents

Importance of Representative Samples

- + What if we only sampled from one factory?
- + Implications for psychological research
 - + Generalizing from sample to population
 - + External validity of research findings
- + Strategies for improving representativeness
 - + Increasing sample size (more M&M packages)
 - + Diversifying sample sources (different stores, batches)
 - + Random selection procedures
 - + Weighting techniques for unequal probability samples

Wrapping Up...

Key Takeaways



1. Sampling is required for understanding populations
2. Larger samples generally provide better estimates
3. Be aware of potential biases in sampling
4. Always aim for representative samples